MILITARY SPECIFICATION

PROJECTOR, RETICLE IMAGE: M14

This amendment forms a part of Military Specification MIL-P-60125A(MU), dated 10 January 1968.

Paragraph 2.1, Drawings: Add "F8565556, Pressure Tester, Two Station".

Paragraph 3.4.2: Delete and substitute:

"3.4.2 Extreme operating temperature.- The projector shall meet the performance requirements of 3.7.1, 3.7.9, 3.7.9.1, 3.7.9.2, 3.7.9.3 and 3.8 while exposed to and thermally stabilized at temperatures of -40° and +150°F."

* Paragraph 3.7.5, Reticle movement: Delete paragraph.

Paragraph 3.10: Add the following sentence:

"The projector shall be sealed with an internal pressure of nitrogen of 3.0 ± .5 psig."

Paragraph 4.2: Delete and substitute:

"4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 reticle projectors plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

* Paragraph 4.3.2.1, TABLE II: In the Requirements column for Characteristic 115, delete "3.7.9.1" and substitute "3.7.7": Delete Characteristic "116 Reticle movement 3.7.5 4.6.7.2.7".

Paragraph 4.4.2, TABLE IV: Delete Characteristics 304, 305, 306, 307, 308, 309, 310, 314, 315, 316 and 317; Delete the "NOTE" at bottom of the table and substitute:

"NOTE: The tests for Characteristics 311, 312, 313 and 318 through 322 inclusive shall be conducted at -40° and +150°F."

Paragraph 4.4.3: Delete and substitute:

"4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3)."

Paragraph 4.5.2.3.4: Delete and substitute:

"4.5.2.3.4 Optical accuracies.- The observation telescope utilized to measure the optical accuracies of the projector shall be at least 10 power (10X) with at least a 1.0 inch entrance pupil. The telescopic system shall be capable of measuring angles in azimuth and elevation with an accuracy of .025 mil. The telescope shall be positioned at infinity focus."

* Paragraph 4.5.3, Item Number 4, Description: Delete "0.30 volts ± 5% maximum." and substitute "0 - 30 volts d.c. ± 5%". Add the following:

"5.	D.C. Power supply	0-10 volts ± 1%
6.	Test circuit	Figure 1
7.	Amplifier	10516404"

Paragraph 4.6.2: Delete and substitute:

"4.6.2 Extreme operating temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The testing equipment shall provide for visual observation of the energized projected reticle pattern during the environmental tests. The projector shall

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be exposed to each temperature extreme specified in 3.4.2 for a period of 5 hours. At the end of each 5 hour period, and while still at the temperature extreme, the projector shall be subjected to the tests in 4.6.7.2.8, 4.6.7.2.9, 4.6.7.2.10 and 4.6.8 through 4.6.12 inclusive. Verification of the motor-tachometer's conformance to the requirements specified in 3.7.9.1, 3.7.9.2 and 3.7.9.3 shall be made by visually observing the movements of the energized reticle, and by monitoring the output voltage of the motor-tachometer. The projector shall conform to the requirements of 3.4.2. Caution shall be exercised during this test to avoid subjecting the projector to thermal shock. The test circuitry utilized to energize, and monitor the output voltages of the projector shall be positioned external to the environmental test chamber."

Paragraph 4.6.5: Delete and substitute:

"4.6.5 Sealing.- The internal pressurizing of the projector to test sealing, shall be performed with special testing equipment conforming to 4.5.2.3.1. Secure the poppet in Valve 10516717 in a closed position. The pressure shall be applied through the input valve in the projector. Pressurize the projector with dry nitrogen gas to the value specified in 3.4.4, in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. The dry nitrogen gas shall have a dew point at least as low as minus 25°F. Retain the applied pressure in the projector for a minimum of 2 hours. Within the first 60 minutes of this 2 hour time period, each elevation and deflection boresight knob shall be rotated through a minimum of 3 full excursions (cw stop - to ccw stop - to cw stop) of their rotational range. Any evidence of leakage in the projector at the end of the 2 hour time period shall not exceed the tolerance specified in 3.4.4, as measured on the pressure gauge of the testing equipment. Upon completion of this test release the poppet in Valve 10516717."

Paragraph 4.6.6: Delete and substitute:

"4.6.6 Interior atmosphere.- Perform this test in sequence to the test in 4.6.5. The special testing equipment specified in 4.6.5 shall be utilized in this test. A dew point tester, with an appropriate adapter to accommodate the Valve 10516717, shall be used to test the dew point of the interior atmosphere of the projector. The projector shall be flushed with dry nitrogen gas. The flushing shall continue until the nitrogen gas bled from the projector has a dew point at least as low as minus 25°F, as measured with the dew point tester. The dew point tester shall be removed from the projector. The internal pressure of nitrogen gas in the projector shall be brought to a measured 3 psig ± 0.5 psig. The internal pressure of the projector shall be monitored for a period of 5 minutes ± 15 seconds. At the end of the 5 minute period there shall be no evidence of leakage in the projector as indicated on the pressure gauge of the testing equipment. The projector shall conform to the requirements of 3.10."

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Paragraph 4.6.7.2.4: Delete "mil graduations on the".

* Paragraph 4.6.7.2.7, Reticle movement: Delete paragraph.

Paragraph 4.6.7.2.11, 5th sentence: Delete "mil scale of the".

Paragraph 4.6.7.2.12, 6th sentence: Delete "mil scale of the".

* Figure 1: Delete and substitute new Figure 1, attached.

*The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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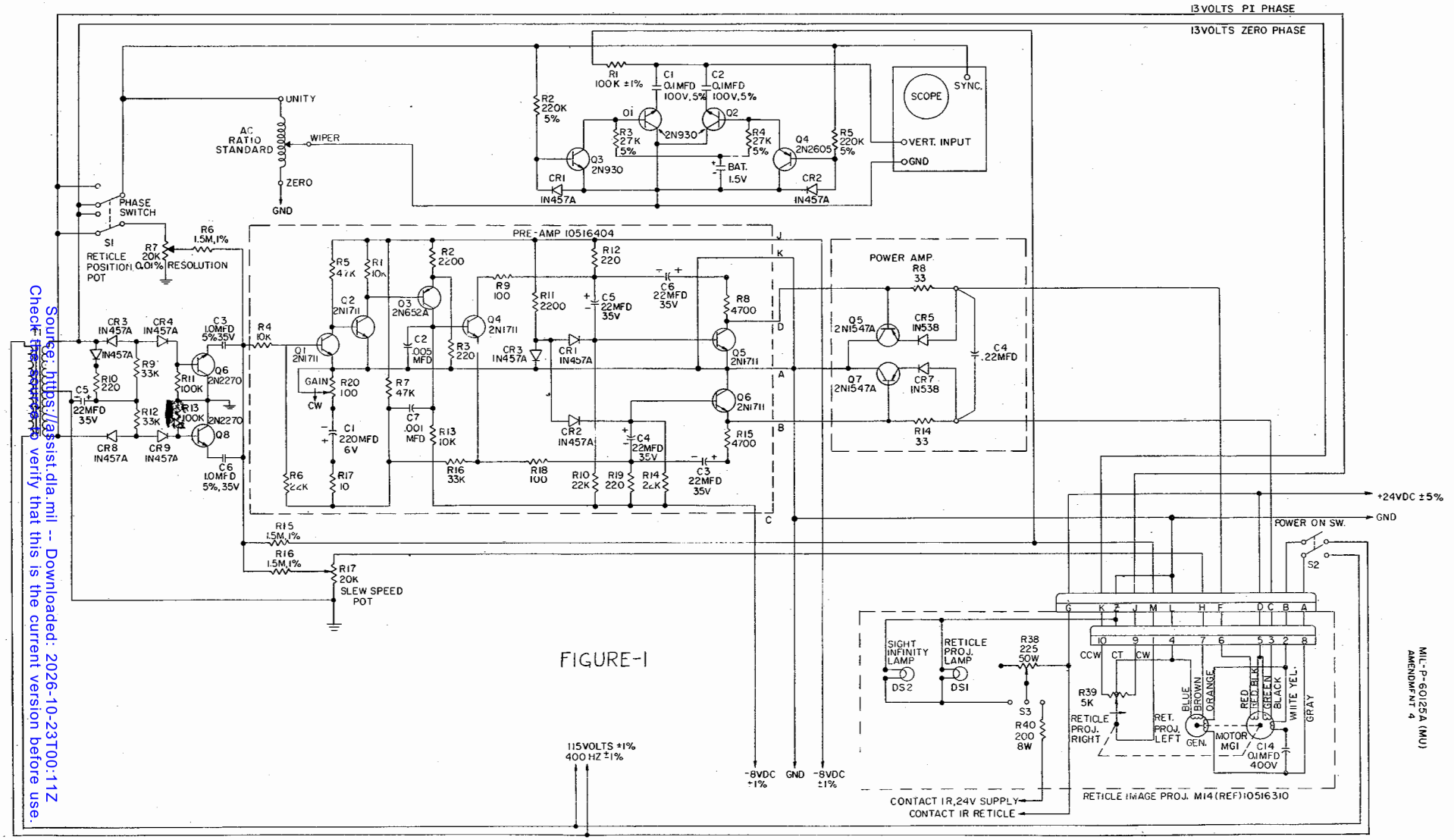
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